

Lab 03 - Motion in Two Dimensions

In this lab, you will use a ball launcher to explore ballistic motion through application of the kinematic equations. First you will launch a ball horizontally and calculate the initial velocity of the projectile. Then you will use the calculated initial velocity to predict the range of the projectile when launched at an incline.

$$x_f = x_i + v_i t + \frac{1}{2} a t^2$$

$$v_f = v_i + a t$$

$$v_f^2 = v_i^2 + 2 a (x_f - x_i)$$

Experiment 1: Horizontal Shot

1. The Ballistic Pendulum must be placed on a table edge. Make sure that the launcher is firing as close as possible to horizontal.
2. Locate a clear line of fire at least 3 meters free of other classmates and other objects.
3. Measure the initial height of the ball from the floor in meters. Take a moment to think where on the ball you should take this measurement to find the overall change in height.

Height ($\Delta y = y_f - y_i$): _____

4. Ready the Ballistic Pendulum for firing and fire a practice shot which will show you how far the ball will travel.
5. Place a piece of white paper on the floor at the approximate impact point. Tape the white paper to the floor and fire the Ballistic Pendulum three times.
6. Measure, in meters, the range of the three shots from where the projectile leaves the firing tube to the impact point. Use the average value in step 7.

x_1 : _____ x_2 : _____ x_3 : _____ x_{ave} : _____

7. Use your knowledge of Kinematics to find the initial velocity in the X and Y direction. Find the initial Vector velocity. Find the final velocity in the X and Y direction. Find the final Vector velocity.

$$x_i =$$

$$y_i =$$

$$x_f =$$

$$y_f =$$

$$v_{ix} =$$

$$v_{iy} =$$

$$v_{fx} =$$

$$v_{fy} =$$

$$a_x =$$

$$a_y =$$

$$t =$$

Experiment 2: Predicting the motion

1. Loosen the bolts on the launcher and reposition the launcher to an incline between 15° and 25° .
2. Measure the vertical distance the ball will drop from the indicated point on the side of the launcher.

Height: _____

3. Calculate the Range the ball will travel using the initial velocity solved for in Part 1.

$x_i =$	$y_i =$
$x_f =$	$y_f =$
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x =$	$a_y =$

$$t =$$

1. Place a piece of paper center at your predicted range and tape it to the ground. Do not gauge the range with a test shot. You have predicted the landing location.
2. **Find the instructor before proceeding with the shot.**
3. Measure, in meters, the range of three shots from where the projectile leaves the firing tube to the impact point. Take the average of the three shots.

x_1 : _____ x_2 : _____ x_3 : _____ x_{ave} : _____

4. Compare your average range to your predicted range.

$$Error = \frac{x_{theory} - x_{experiment}}{x_{theory}} * 100$$

Percent Error: _____

Questions:

1. Would knowing the radius of the ball refine your value for initial velocity?
2. Would knowing the mass of the ball improve your calculations?